

The Equation.

See it. Name it. Say it. Use it.

A working book on logic, questions, quality and control.

G = P × M × E × D

Five letters. One rule inside them: **a zero anywhere is a zero everywhere.** By the last page you will read this line the way you read $2 \times 2 = 4$: at a glance, without anyone translating it for you.

PREPARED FOR

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FROM THE DESK OF

Chairman Vyc

OCTOBER 3, 2026

How every THYLORA equation is taught.

Words alone, you forget. Symbols alone, you can't read. You need both, and you need to move back and forth between them. Every equation in this book goes through the same four steps, in the same order.

1

SEE IT

The full formula, written out as math. No shortcuts and nothing left out.

2

NAME IT

Every letter and symbol gets defined. Nothing stays a mystery.

3

SAY IT

The expression as one plain sentence you could say out loud.

4

USE IT

Put real numbers in. Work it to the end. Then use it on a new problem.

WHY IT WORKS · THE 2 × 2 TEST

$$2 \times 2 = 4$$

Nobody reads that as “two geese, times two more geese, equals four geese.” You **see** it and you **know** it. The shape carries the meaning, so the meaning can be anything: geese, dollars, hours.

That is the goal for every long equation in our system. After enough practice you look at $G = P \times M \times E \times D$ and think: “*every part has to be there, or nothing comes out.*”

THE RULE BEHIND THE RULE · ON RECORD

$$M_c = W \times F \times V \times Q \times Q_2 \times E \times X$$

M_c	math communication is complete
W	the words / plain meaning are there
F	the full formula is visibly there
V	every variable is defined
Q	the direct question is stated
Q_2	the deeper question behind it is stated
E	the evidence it needs is stated
X	a worked example is shown

THYLORA registry MATH-DISPLAY-CLOSURE-625. If the formula is missing, $F = 0$, so the whole explanation scores zero.

This book holds itself to that rule. **Every equation in it shows the formula, names every symbol, says it in words, and works an example.** If you ever find one that doesn't, that's a defect. Report it.

$$G = P \times M \times E \times D$$

$$G = P \times M \times E \times D$$

STEP 1 · SEE IT

STEP 2 · NAME IT PROPOSED READING · CHAIRMAN CONFIRMS

G	Goal reached	How fully the result you wanted actually happened. This is the output.
P	Purpose	A clear "why" and a clear target. If nobody can say what done looks like, P = 0.
M	Method	A real way to do it: steps, tools, people, a path.
E	Evidence	Proof that it's working: something measured, seen, or recorded, not just believed.
D	Discipline	Doing it again tomorrow. Follow-through, consistency, finishing.
×	Times	Multiply. Each part scales the others. It is not added on top.

STEP 3 · SAY IT

"How far you get equals your **purpose**, times your **method**, times your **evidence**, times your **discipline**. If any one of them is zero, you get nowhere, no matter how strong the others are."

Why multiply and not add?

With adding, a missing piece gets hidden. With multiplying, it gets exposed. A great purpose, a great method and great discipline still produce **zero** if there's no evidence you're on target. That is how real work behaves, so the equation is written to behave that way too.

How to score it

Rate each factor from 0 to 5.

0 = absent · 3 = usable · 5 = locked and proven.

Highest possible score: $5 \times 5 \times 5 \times 5 = 5^4 = 625$.

Any single 0 means **HOLD**: fix that factor first.

Put it together.

Example: a report you are getting ready for Monday. You have a strong purpose, a good method and steady habits, but nobody has checked the numbers yet.

$$G = P \times M \times E \times D$$

$$G = 5 \times 4 \times 0 \times 4$$

$$G = 0 \text{ out of 625. HOLD. The evidence is missing.}$$

Now someone checks the numbers against the source, which brings E up to a 3:

$$G = 5 \times 4 \times 3 \times 4$$

$$G = 20 \times 3 \times 4 = 60 \times 4$$

$$G = 240 \text{ out of 625} = 38.4\%. \text{ Moving. Next, raise the lowest factor.}$$

THE TRAP ADDITION SETS

$$5 + 4 + 0 + 4 = 13 \text{ of } 20 = 65\%$$

Adding the same scores reports 65%, which looks fine. It hides the zero. **That is how broken work gets labeled "almost done."**

FIX THE SMALLEST FACTOR FIRST

From $5 \times 4 \times 3 \times 4$: raising E from 3 to 4 adds $5 \times 4 \times 1 \times 4 = 80$. Raising P from 5 to 6 isn't possible, since 5 is the top. **With multiplication, raising your weakest link is almost always the move that pays the most.**

SAME SHAPE, ANY MEANING · SWAP THE LETTERS

$$\text{Pancake} = P \times M \times E \times D$$

P = pancake powder · **M** = milk · **E** = eggs · **D** = done-heat. No eggs, no pancake. A great pan doesn't save you. You already knew this. Now you can *see* it.

$$\text{Arrive} = P \times M \times E \times D$$

P = plan / address · **M** = machine (the car) · **E** = energy (gas) · **D** = driver. Empty tank: $E = 0$, so $\text{Arrive} = 0$. Same equation, new problem. That is transfer.

Quality is a distance. Control is closing it.

People say “quality control” like it's one word. It's two ideas, and each one has its own equation.

$$Q = 1 - |R - S| \div S$$

QUALITY

$$x_{n+1} = x_n + k(S - x_n)$$

CONTROL

Q	Quality	1.00 = exactly on standard
R	Result	what you actually got
S	Standard	what it was supposed to be
 	Distance	how far off, ignoring over or under

Say it: “Quality is how close the result came to the standard.”

$S = 12 \text{ min bake} \cdot R = 15 \text{ min}$
 $Q = 1 - |15 - 12| \div 12 = 1 - 3 \div 12$
 $Q = 1 - 0.25 = 0.75$

x_n	Now	where it stands this round
x_{n+1}	Next	where it stands after you correct
$S -$	The gap	distance left to the standard
x_n		
k	Correction	how much of the gap you close each time (0 to 1)

Say it: “Each round, measure the gap and close part of it.”

$S = 100 \cdot x_0 = 80 \cdot k = 0.5$
 $x_1 = 80 + 0.5(20) = 90$
 $x_2 = 90 + 0.5(10) = 95 \rightarrow x_3 = 97.5$

THE LOOP, IN FOUR WORDS

Measure → Compare → Correct → Record. Skip “record” and the next person starts over from zero. That's how a system forgets.

Chairman's version: “Less distance between me and the snacks.” He's right, and it's the same equation. $S - x_n$ is the distance. Control is making it smaller every round.

Between deduction and conclusion, something leaks.

A **deduction** is a chain: if these things are true, this must follow. A **conclusion** is what a person decides to believe and act on. People treat them as the same thing. They aren't, and the difference between them can be measured.

$$S = V \times p_1 \times p_2 \times \dots \times p_n \quad \Delta = C - S$$

SUPPORTED CERTAINTY · AND THE GAP

S	Supported	how sure the evidence actually lets you be (0 to 1)
V	Valid form	1 if the conclusion really follows from the premises, 0 if the logic is broken
p_i	Premise i	the chance that premise number <i>i</i> is actually true (0 to 1)
C	Claimed	how sure the person sounds when they state the conclusion
Δ	The gap	overconfidence: claimed certainty minus supported certainty

Say it: “Your conclusion can only be as strong as all of your premises together. Every premise you're unsure of weakens it.”
(This multiplies when the premises are independent of each other. When they depend on each other, use their joint probability instead. The lesson is the same.)

Three premises, each 90% sure, with perfect logic. The person says it's “definitely true.”

$$S = 1 \times 0.9 \times 0.9 \times 0.9 = \mathbf{0.729}$$

$$\Delta = 1.00 - 0.729 = \mathbf{0.271} \quad \text{27\% of the certainty was never earned.}$$

Claimed, C = 1.00

Supported, S = 0.729



With **10** premises at 90% each: $0.9^{10} \approx 0.35$. A long chain of “pretty sure” steps ends at “probably not.”

Five places the conclusion outruns the proof.

LEAK	WHAT HAPPENS	EXAMPLE	ASK THIS
False premise	The logic is fine but one of its starting points is wrong. Valid, but not sound.	"All our suppliers are reliable. Ace is our supplier. So Ace is reliable." What if one supplier isn't?	Is each premise actually checked?
Hidden premise	A step nobody said out loud is carrying the whole argument.	"Sales fell, so the ad failed." Hidden premise: nothing else changed.	What am I assuming without saying it?
Overreach	The conclusion claims more than the premises cover.	"Three customers loved it, so everyone will."	Does the conclusion stay inside the evidence?
Backwards "if"	Affirming the consequent: "If A then B. B happened. So A."	"If it rained, the street is wet. The street is wet. So it rained." A hydrant could have done it.	What else could produce B?
Shifting word	A word quietly changes its meaning partway through the argument.	"Key is light on her feet. Light things float."	Does every word mean the same thing in every step?

FAMOUS MISLABEL

Sherlock Holmes doesn't deduce.

When Holmes looks at a tan and a posture and says "you've been in Afghanistan," that isn't deduction. It's **abduction**: picking the best explanation. It can be wrong. It feels certain because it's fast. **Fast isn't the same as certain.**

THE GAP RULE · FOR EVERY REPORT

$C \leq S$

Never state a conclusion more firmly than the evidence supports. When the evidence is 0.73, say "likely," not "certain." Then name the weakest premise. Strengthening that premise is your next job.

CHAIRMAN'S METHOD, WRITTEN AS MATH

"What question must be answered before the next question is even useful?" In the equation, that question is **the lowest p_i** . Answer it first, and S rises the most for the least work. It's the same move as fixing the smallest factor in $G = P \times M \times E \times D$.

Eleven kinds of logic.

Every culture reasons. They start from different places. Here are the main tools, each with its formula, what it's good for, and where it breaks.

LOGIC	SHAPE	WHAT IT DOES	WHERE IT BREAKS
Deductive	$A \rightarrow B, A \vdash B$	Certain, if the premises are true.	Can't create new facts, and inherits any false premise.
Inductive	$x_1 \dots x_n \Rightarrow \forall x$	Learns a pattern from examples.	The next case can break the pattern.
Abductive	B; A best explains B	Picks the best explanation. Detectives, doctors, mechanics.	The best story isn't always the true one.
Analogical	$A : B :: C : ?$	Moves understanding from a known problem to a new one.	The two things may only look alike.
Bayesian	$P(H E) = P(E H) \cdot P(H) \div P(E)$	Updates belief as evidence comes in.	A bad starting belief $P(H)$ skews everything after it.
Fuzzy	$\text{truth} \in [0, 1]$	Things can be partly true. Warm, not hot. (Zadeh, 1965)	The cutoffs are a judgment call.
Dialectical	thesis $\rightleftharpoons$ antithesis $\rightarrow$ synthesis	Puts two opposing views together to find something better.	Can turn into arguing for its own sake.
Four-cornered	T, F, both, neither	Nāgārjuna (India, about 150 CE): allows answers outside yes/no.	Hard to act on without a decision rule.
Five-step Nyāya	claim $\rightarrow$ reason $\rightarrow$ example $\rightarrow$ apply $\rightarrow$ conclude	Indian method that requires a real-world example in the middle of the argument.	Slower, which is the point.
Erotetic	$Q \Rightarrow Q'$	The logic of questions : which questions validly lead to which other questions.	Most people never learn it exists.
Relational (Ubuntu)	$\text{Self} = f(\text{Us})$	Truth and identity are judged through relationships, not alone.	Can be ignored by systems built around a single individual.

Erotetic is the word you used, Chairman. It names your method: *a question produces the next question*. Your registry already writes it as $Q = f(K, E, C)$: the question you can ask is a function of Knowledge, Evidence and Connections.

Nine ways to ask a better question.

You ask questions in a particular way. These are other ways. Each one is a tool. Pick the one that fits the problem.

Socratic

Keep asking “what do you mean by that?” until the definition holds up.

“What would make that false?”

Five Whys

Toyota's method: ask “why?” five times and you get from the symptom to the cause.

why⁵(problem) → root

Sankofa

Akan, from Ghana: go back and fetch what was left behind before you move forward.

“What did we already know?”

Inversion

“Invert, always invert” (Jacobi). Ask how to guarantee failure, then avoid those things.

“How would this fail?”

Pre-mortem

Pretend it's already a year from now and the project failed. Write down why.

“It failed. What killed it?”

Steelman

State the other side's best case better than they can, then answer that.

“What's their strongest point?”

First principles

Strip the problem down to what you know for certain, then build back up.

“What is true no matter what?”

Counterfactual

Change one thing in your head and see what moves.

“If X were gone, what changes?”

The Chairman's question

Look past the next gate. Verify backward, then move forward.

“What must be answered before this is useful?”

HOW TO CHOOSE

If you're stuck on a definition, go **Socratic**. If the same problem keeps coming back, use **Five Whys**. If you're about to launch something, run a **pre-mortem**. If you're in a disagreement, **steelman** the other side. If something got lost along the way, go back to it: **Sankofa**. If you don't know where to start, use the Chairman's question.

Africa was writing equations 3,500 years ago.

Not a slogan: the record. Each claim is with each claim marked **VERIFIED**, **DEBATED**, or **BIAS ON RECORD**.

EGYPT · AHMES (RHIND) PAPYRUS · COPIED ABOUT 1550 BCE FROM AN OLDER TEXT **VERIFIED**

Problem 24: "A quantity and its seventh, added together, become 19. What is the quantity?" That is an algebra problem, written roughly 3,500 years ago. The scribe solved it with **false position**: guess, check, then scale.

$$x + x \div 7 = 19 \quad \text{See it}$$

Guess $x = 7$: $7 + 1 = 8$ wrong, but it shows the shape

$$\text{Scale: } 19 \div 8 = 2\frac{3}{8} \rightarrow x = 7 \times 2\frac{3}{8} = 16\frac{1}{2} \quad \text{Check: } 16\frac{1}{2} + 2\frac{3}{8} = 19 \checkmark$$

The papyrus is held at the British Museum. Ahmes says he copied it from an earlier work, from about 1850 BCE.

YORUBA · IFÁ **VERIFIED**

$$16 \times 16 = 2^8 = 256$$

Ifá divination reads marks that are either single or double. That's binary, two states. Eight marks give 256 *odù*, each tied to verses of knowledge. **DEBATED** Ron Eglash argues this binary passed through Arab geomancy into Europe and on to Leibniz, the father of binary code. Leibniz himself credited the Chinese I Ching. The link is argued, not proven.

EGYPT · MA'AT **VERIFIED**

$$\text{Heart} \leq \text{Feather}$$

In the Book of the Dead (Spell 125), the heart is weighed against the feather of Ma'at: truth, balance, order. The soul speaks 42 declarations of what it did not do. Truth is shown as a **balance**, an equation held level. If the heart is heavier, you fail.

CONGO · ISHANGO BONE, ABOUT 20,000 YEARS OLD **VERIFIED** **DEBATED**

One of the oldest counting objects we know of. One column is notched 11, 13, 17, 19, which are all the prime numbers between 10 and 20. Whether that's deliberate math or coincidence is still argued. That it is deliberate counting is not.

ANGOLA · CHOKWE SONA **VERIFIED**

Sand drawings traced in a single continuous line around a grid of dots, without lifting the finger. That is the same problem Euler made famous in 1736 (the Königsberg bridges). Documented by mathematician Paulus Gerdes.

European logic and African logic.

You asked this and it went unanswered. Here's the answer: **logic is not white or Black. Starting points differ.** And for a long time the record was written by people who claimed Africans had no logic at all.

TWO STARTING POINTS

Self = f(Thought)

Descartes, 1637: *"I think, therefore I am."* You start alone, inside your own head.

Self = f(Us)

John Mbiti, 1969, describing African thought: *"I am, because we are; and since we are, therefore I am."* You start in relationship. Zulu: *Umuntu ngumuntu ngabantu*, a person is a person through other people.

BIAS ON RECORD VERIFIED QUOTES

Hume, 1753: footnote, "I am apt to suspect the negroes ... to be naturally inferior to the whites."

Kant, 1764: wrote that Africans have "by nature no feeling that rises above the trifling."

Hegel, 1830s: called Africa "no historical part of the World."

Lévy-Bruhl, 1910: called non-European minds "prelogical." He abandoned that idea in his own late notebooks, published in 1949. Most textbooks still quote the 1910 version.

HIDDEN NARRATIVE · THE STRONGEST ARGUMENTS ON EACH SIDE

Greek debt to Egypt

Greek writers themselves (Herodotus; Plato in *Timaeus*) credit Egyptian priests. G.G.M. James, *Stolen Legacy* (1954), says Greek philosophy was taken from Egypt. Martin Bernal, *Black Athena* (1987), shows 19th-century scholars deliberately erased that credit. **DEBATED** Mary Lefkowitz, *Not Out of Africa* (1996), disputes "stolen." The influence is real. How much is still argued.

Who knew the land

Your point about 1923 has scholarship behind it. Judith Carney, *Black Rice* (2001): Carolina rice planting depended on West African knowledge brought by enslaved people. **DEBATED** Eltis, Morgan & Richardson (2007) argue planters drove the system. Both sides agree African labor and skill were central.

Ethiopia, 1667

Zera Yacob's *Hatata* argues for reasoning things out yourself instead of accepting authority, 170 years before Hegel. **DEBATED** Some scholars say a 19th-century Italian monk wrote it. Others defend Zera Yacob as the author. It is still open.

The equation for this whole page: Truth = Record – Bias. Read the original document, name who wrote the commentary on it, and keep the parts that survive both.

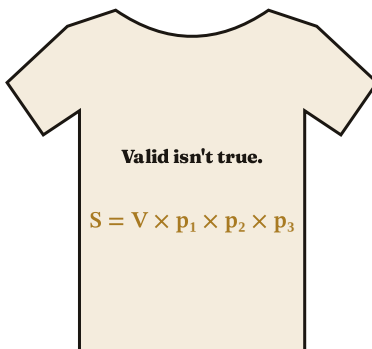
A saying. The equation. What it means.

Most shirts print a quote that doesn't mean much. Ours print what we mean, then the math that proves it, then a short explanation on the back. Somebody reads it, understands it, and starts asking better questions.



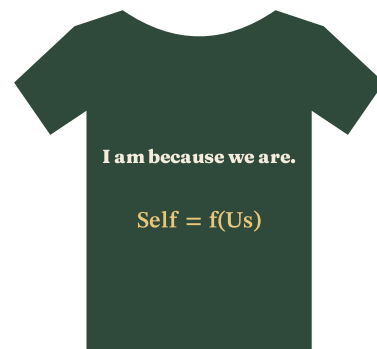
01 · BLACK / GOLD

Back: Purpose, Method, Evidence, Discipline get multiplied, not added. Miss one and the goal is zero.



02 · CREAM / INK

Back: Good logic built on weak premises gives a weak conclusion. Three 90% premises leave you 73% sure.



03 · FOREST / GOLD

Back: Ubuntu. Who you are is a function of your people. Descartes started alone. We start together.



04 · SAND / RED

Back: Ma'at, Egypt. Truth was pictured as a balance. Your heart cannot outweigh the truth.



05 · NAVY / GOLD

Back: The question you can ask grows with your Knowledge, your Evidence and your Connections.



06 · BLACK / GOLD

Back: Rhind Papyrus, Problem 24, Egypt, about 1550 BCE. $x = 16\%$. Look it up.

Production note: shirt 06's claim is accurate for written algebra problems. Babylonian algebra tablets are about as old, so the claim is "before Greece," not "before everyone." Before printing, each shirt goes through Famous Thought Verification, $F = S \times A \times C \times T$ (source, attribution, context, transfer).

Three exercises. Pencil.

1 • Score something real you're working on

$$G = P \times M \times E \times D = ___ \times ___ \times ___ \times ___ = ______ / 625$$

Which factor is lowest? _____ What would raise it by 1 this week?

2 • Find the gap

"Our last two posts did well on Tuesday. Tuesday is our best day. We should post only on Tuesdays."

Premises: (1) _____ $p_1 = ___$ (2) _____ $p_2 = ___$

$$S = V \times p_1 \times p_2 = ______ \quad \Delta = C - S = ______$$

Which leak is this (page 7)? _____

3 • Ask the question before the question

Problem: "Why aren't the store products selling?"

Before that question is useful, what must be answered first? Try two methods from page 9.

A NOTE FROM THE CHAIRMAN'S DESK

You used to see it fast and understand it fast. That speed is still in you. These equations don't replace your logic. They give it something to hold so it doesn't drift. See it. Name it. Say it. Use it.

Answer key for 2: the main leak is **overreach** (two posts can't stand for every week), with a hidden premise (nothing else made those posts do well). If you rated the premises 0.6 and 0.5: $S = 1 \times 0.6 \times 0.5 = 0.30$. Stated as "we should," $C \approx 0.9$, so $\Delta \approx 0.60$.